SECURITY INCIDENT REPORT

Exercise source: malware-traffic-analysis.net | File: 2026-02-28-traffic-analysis-exercise.pcap

Report Date	2026-02-28	**Severity**	**HIGH**
Incident Time (UTC)	19:55:51 UTC	**Status**	**UNDER INVESTIGATION**
Domain	easyas123.tech	**Exercise Source**	malware-traffic-analysis.net

1. Executive Summary

As part of a Security Operations Center (SOC) exercise from malware-traffic-analysis.net, the SIEM system generated multiple signature alerts for NetSupport Manager RAT activity originating from an internal host communicating with external C2 server 45.131.214[.]85 over TCP port 443. The activity started on 2026-02-28 at 19:55 UTC.

Packet capture (pcap) analysis confirmed that Windows workstation DESKTOP-TEYQ2NR, assigned to user Brad Rolf (account: brolf), was actively infected and beaconing to the attacker-controlled C2 server using the NetSupport Manager RAT protocol. The host must be isolated immediately and subject to forensic examination.

2. Infected Host — Key Findings

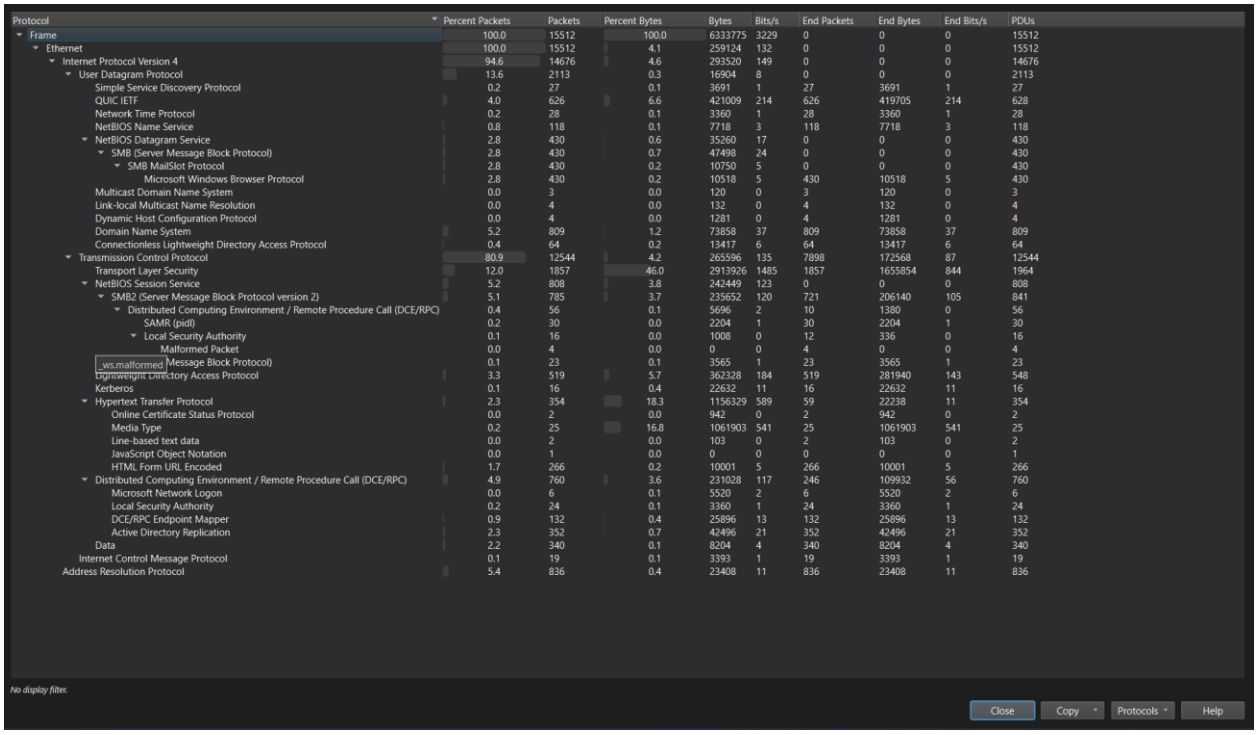
FIELD	VALUE
IP Address	`10.2.28.88`
MAC Address	`00:19:d1:b2:4d:ad`
Hostname	`DESKTOP-TEYQ2NR`
AD Domain	`EASYAS123 / easyas123.tech`
User Account	`brolf`
Full Name	`Brad Rolf`
C2 Server	`45.131.214[.]85 (TCP port 443)`
Malware	`NetSupport Manager RAT v1.3`
First Seen (UTC)	`2026-02-28 19:55:51`

3. Evidence — Step-by-Step Analysis

3.1 Protocol Hierarchy Overview

Opening the pcap in Wireshark and navigating to Statistics → Protocol Hierarchy gives an immediate overview of all protocol activity in the capture. Key protocols visible include HTTP (NetSupport RAT beacons), Kerberos (user authentication), NBNS (hostname broadcasts), and DHCP (IP assignment).

Wireshark path: `Statistics → Protocol Hierarchy`



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	15512	100.0	6333775	3229	0	0	0	15512
Ethernet	100.0	15512	4.1	259124	132	0	0	0	15512
Internet Protocol Version 4	94.6	14676	4.6	293520	149	0	0	0	14676
User Datagram Protocol	13.6	2113	0.3	16904	8	0	0	0	2113
Simple Service Discovery Protocol	0.2	27	0.1	3691	1	27	3691	1	27
QUIC IETF	4.0	626	6.6	421009	214	626	419705	214	628
Network Time Protocol	0.2	28	0.1	3360	1	28	3360	1	28
NetBIOS Name Service	0.8	118	0.1	7718	3	118	7718	3	118
NetBIOS Datagram Service	2.8	430	0.6	35260	17	0	0	0	430
SMB (Server Message Block Protocol)	2.8	430	0.7	47498	24	0	0	0	430
SMB MailSlot Protocol	2.8	430	0.2	10750	5	0	0	0	430
Microsoft Windows Browser Protocol	2.8	430	0.2	10518	5	430	10518	5	430
Multicast Domain Name System	0.0	3	0.0	120	0	3	120	0	3
Link-local Multicast Name Resolution	0.0	4	0.0	132	0	4	132	0	4
Dynamic Host Configuration Protocol	0.0	4	0.0	1281	0	4	1281	0	4
Domain Name System	5.2	809	1.2	73858	37	809	73858	37	809
Connectionless Lightweight Directory Access Protocol	0.4	64	0.2	13417	6	64	13417	6	64
Transmission Control Protocol	80.9	12544	4.2	265596	135	7898	172568	87	12544
Transport Layer Security	12.0	1857	46.0	2913926	1485	1857	1655854	844	1964
NetBIOS Session Service	5.2	808	3.8	242449	123	0	0	0	808
SMB2 (Server Message Block Protocol version 2)	5.1	785	3.7	235652	120	721	206140	105	841
Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	0.4	56	0.1	5696	2	10	1380	0	56
SAMR (pidl)	0.2	30	0.0	2204	1	30	2204	1	30
Local Security Authority	0.1	16	0.0	1008	0	12	336	0	16
Malformed Packet	0.0	4	0.0	0	0	4	0	0	4
_ws.malformed Message Block Protocol)	0.1	23	0.1	3565	1	23	3565	1	23
Lightweight Directory Access Protocol	3.3	519	5.7	362328	184	519	281940	143	548
Kerberos	0.1	16	0.4	22632	11	16	22632	11	16
Hypertext Transfer Protocol	2.3	354	18.3	1156329	589	59	22238	11	354
Online Certificate Status Protocol	0.0	2	0.0	942	0	2	942	0	2
Media Type	0.2	25	16.8	1061903	541	25	1061903	541	25
Line-based text data	0.0	2	0.0	103	0	2	103	0	2
JavaScript Object Notation	0.0	1	0.0	0	0	0	0	0	1
HTML Form URL Encoded	1.7	266	0.2	10001	5	266	10001	5	266
Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	4.9	760	3.6	231028	117	246	109932	56	760
Microsoft Network Logon	0.0	6	0.1	5520	2	6	5520	2	6
Local Security Authority	0.2	24	0.1	3360	1	24	3360	1	24
DCE/RPC Endpoint Mapper	0.9	132	0.4	25896	13	132	25896	13	132
Active Directory Replication	2.3	352	0.7	42496	21	352	42496	21	352
Data	2.2	340	0.1	8204	4	340	8204	4	340
Internet Control Message Protocol	0.1	19	0.1	3393	1	19	3393	1	19
Address Resolution Protocol	5.4	836	0.4	23408	11	836	23408	11	836

Figure: Wireshark — Protocol Hierarchy showing HTTP, Kerberos, NBNS and DHCP traffic

3.2 Network Conversations — Identifying the Infected Host

Statistics → Conversations (IPv4 tab) reveals all unique IP pairs in the capture. The host 10.2.28.88 has the most significant external conversation volume, and the connection to 45.131.214.85 with sustained traffic confirms this is the infected machine communicating with the C2 server.

Wireshark path: `Statistics → Conversations → IPv4 tab`

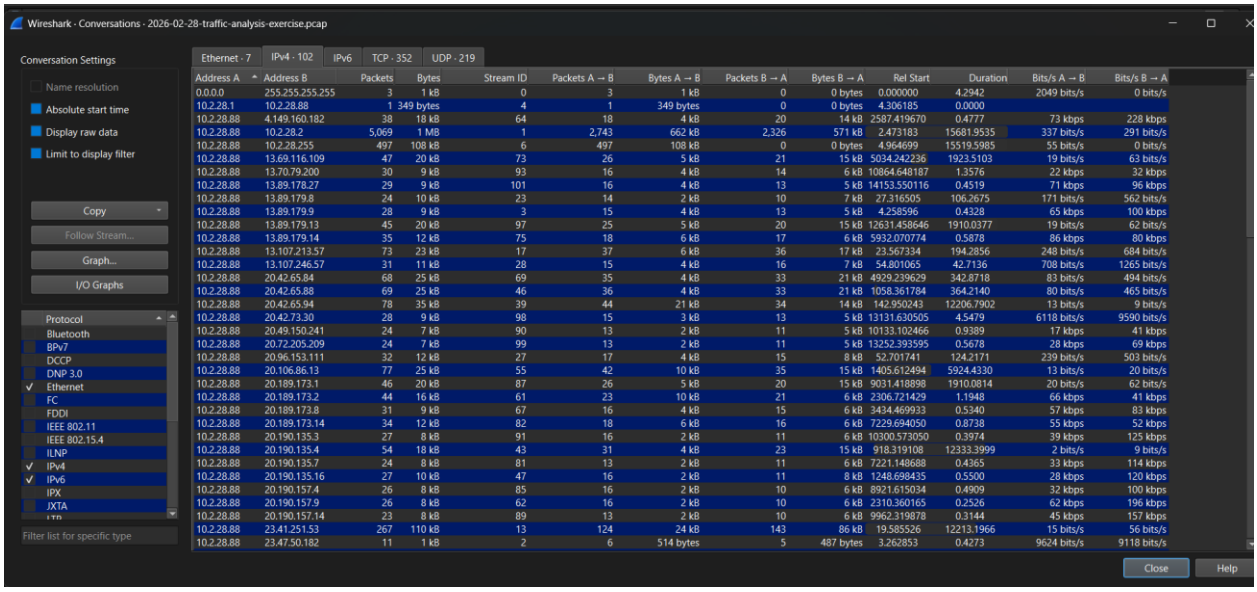


Figure: Wireshark — Conversations view showing 10.2.28.88 as the primary communicating host

3.3 IP Address, MAC Address & Hostname (DHCP)

Filtering for DHCP traffic isolates the IP assignment exchange. The DHCP Request packet (frame 103) reveals the MAC address in the Ethernet II header, the requested IP address in Option 50, and the machine hostname in Option 12.

Wireshark filter: `dhcp`

- MAC Address: 00:19:d1:b2:4d:ad — found in Ethernet II → Source
- IP Address: 10.2.28.88 — found in Option (50) Requested IP Address
- Hostname: DESKTOP-TEYQ2NR — found in Option (12) Host Name

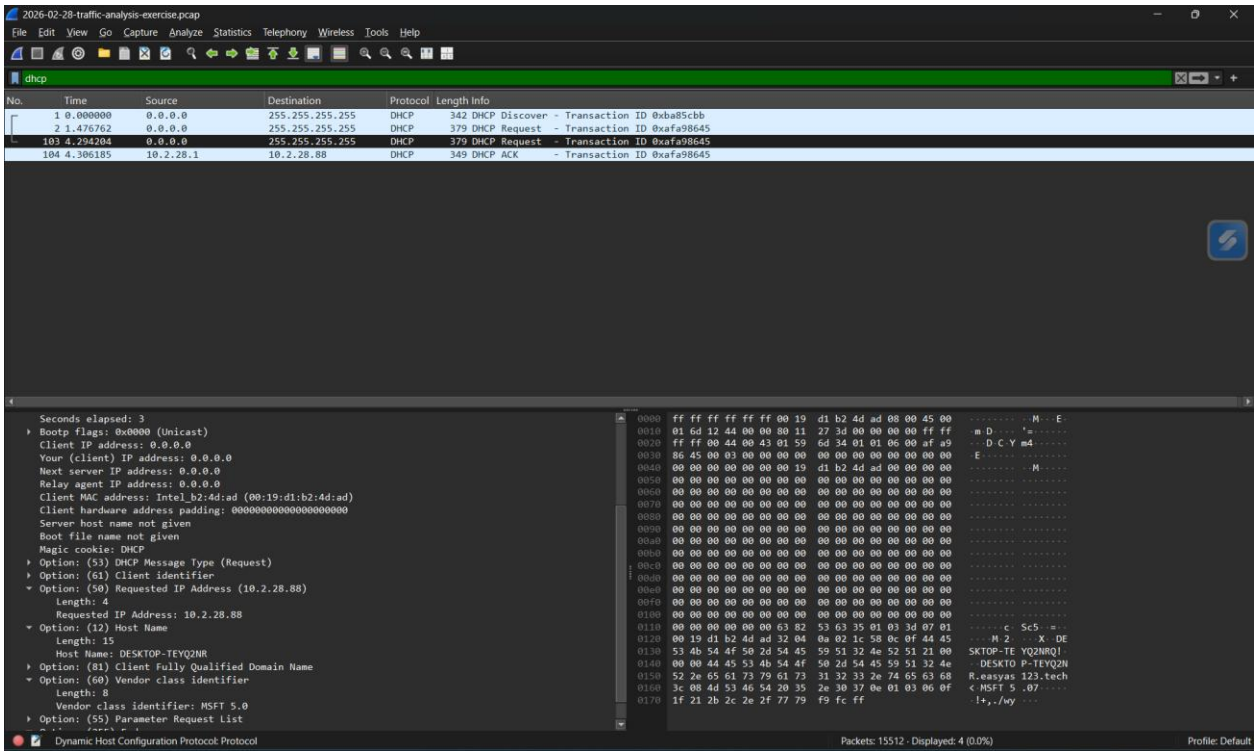


Figure: Wireshark — DHCP Request (filter: dhcp) showing MAC 00:19:d1:b2:4d:ad, IP 10.2.28.88, and hostname DESKTOP-TEYQ2NR

3.4 Hostname Confirmation (NetBIOS Name Service)

NBNS registration broadcasts from 10.2.28.88 independently confirm the hostname DESKTOP-TEYQ2NR. These packets are sent at startup as the machine announces its NetBIOS name to the LAN.

Wireshark filter: `nbns and ip.src == 10.2.28.88`

- Hostname confirmed: DESKTOP-TEYQ2NR — in NetBIOS Name Service → Additional records → Name

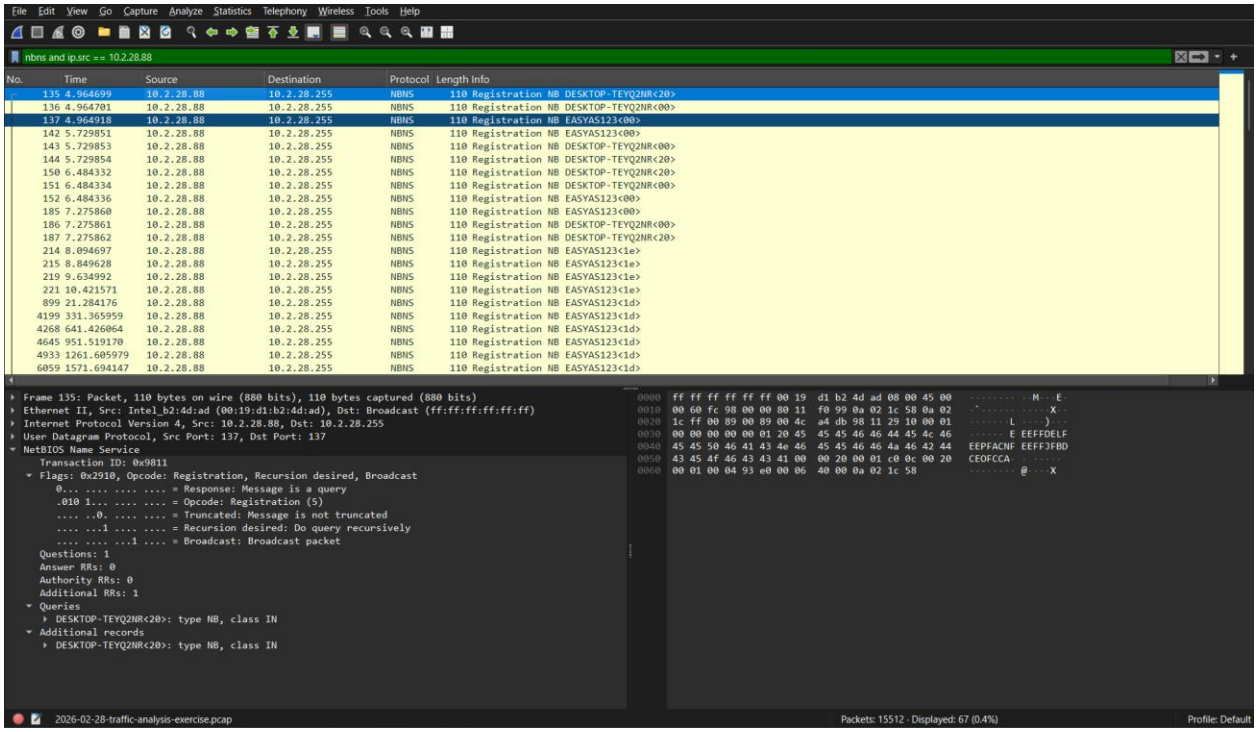


Figure: Wireshark — NBNS registration packets from 10.2.28.88 confirming hostname DESKTOP-TEYQ2NR

3.5 User Account Name (Kerberos Authentication)

Filtering for Kerberos traffic and examining the TGS-REP packets reveals the CNameString field, which contains the Active Directory username of the authenticated user on the infected machine.

Wireshark filter: `kerberos.CNameString`

- Kerberos packet expanded: Kerberos → tgs-rep → cname → CNameString: brolf
- Realm: EASYAS123.TECH — confirming Active Directory domain membership

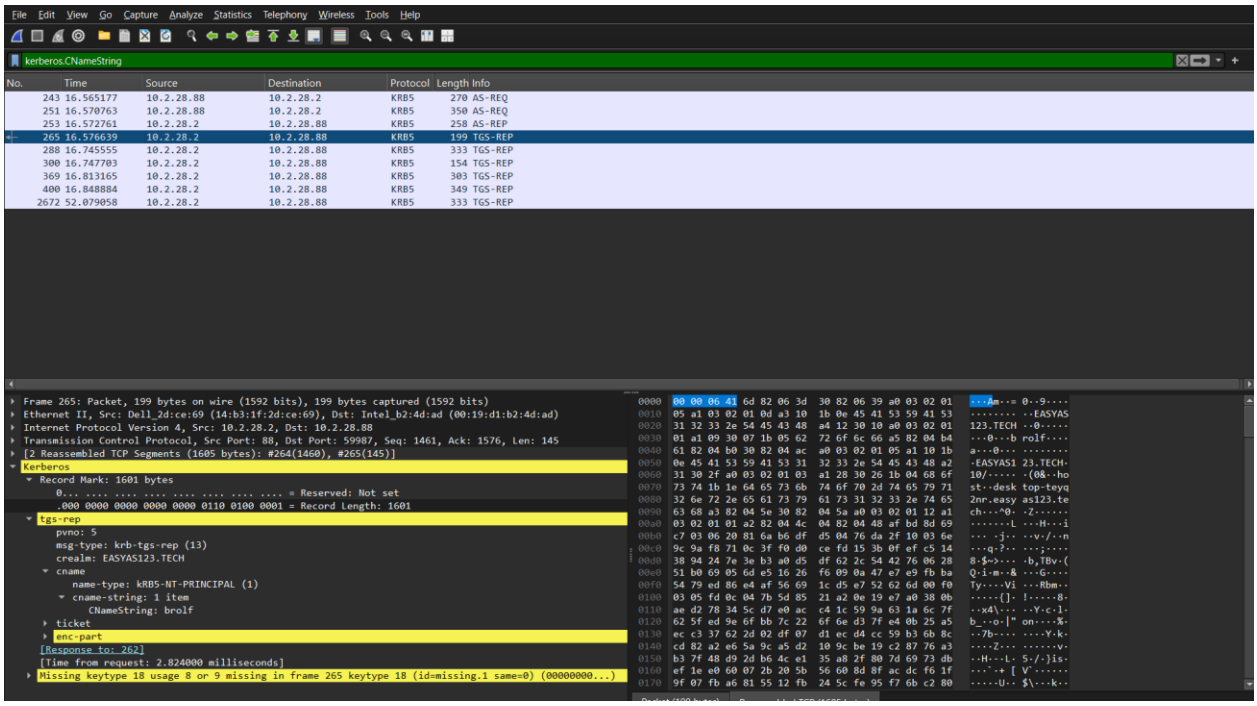


Figure: Wireshark — Kerberos TGS-REP (filter: kerberos.CNameString) showing CNameString: brolf and realm EASYAS123.TECH

3.6 NetSupport Manager RAT C2 Beacon (HTTP POST)

Filtering for HTTP POST requests to the C2 server shows the RAT's repeated beacon loop. The very first POST (frame 2638) is in plaintext — the User-Agent header identifies the malware as NetSupport Manager v1.3, and the POST body contains the initial check-in command CMD=POLL.

Wireshark filter: `http.request.method == POST and ip.dst == 45.131.214.85`

- User-Agent: NetSupport Manager/1.3 (malware identification)
- POST URL: http://45.131.214.85/fakeurl.htm
- Beacon payload (frame 2638): CMD=POLL\nINFO=1\nACK=1 (plaintext)
- Subsequent payloads: CMD=ENCD (XOR-encrypted C2 commands)

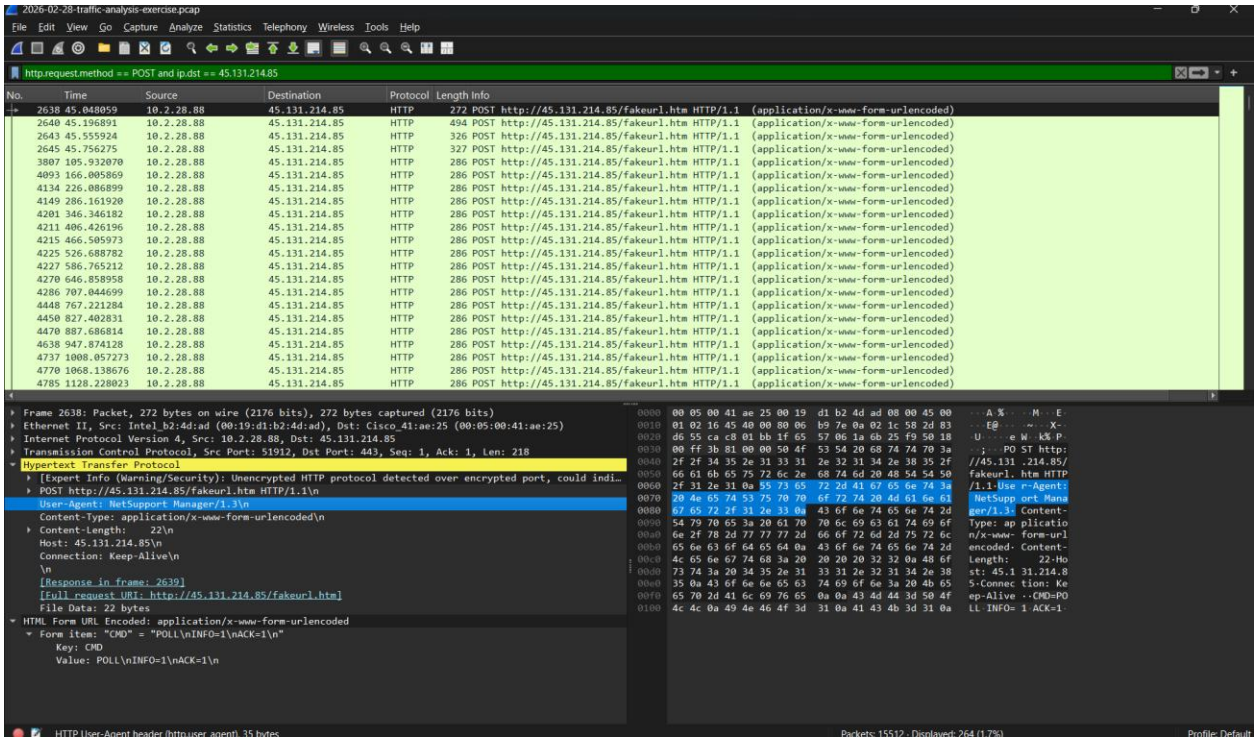


Figure: Wireshark — HTTP POST filter showing all RAT beacons to fakeurl.htm with User-Agent: NetSupport Manager/1.3 and CMD=POLL payload

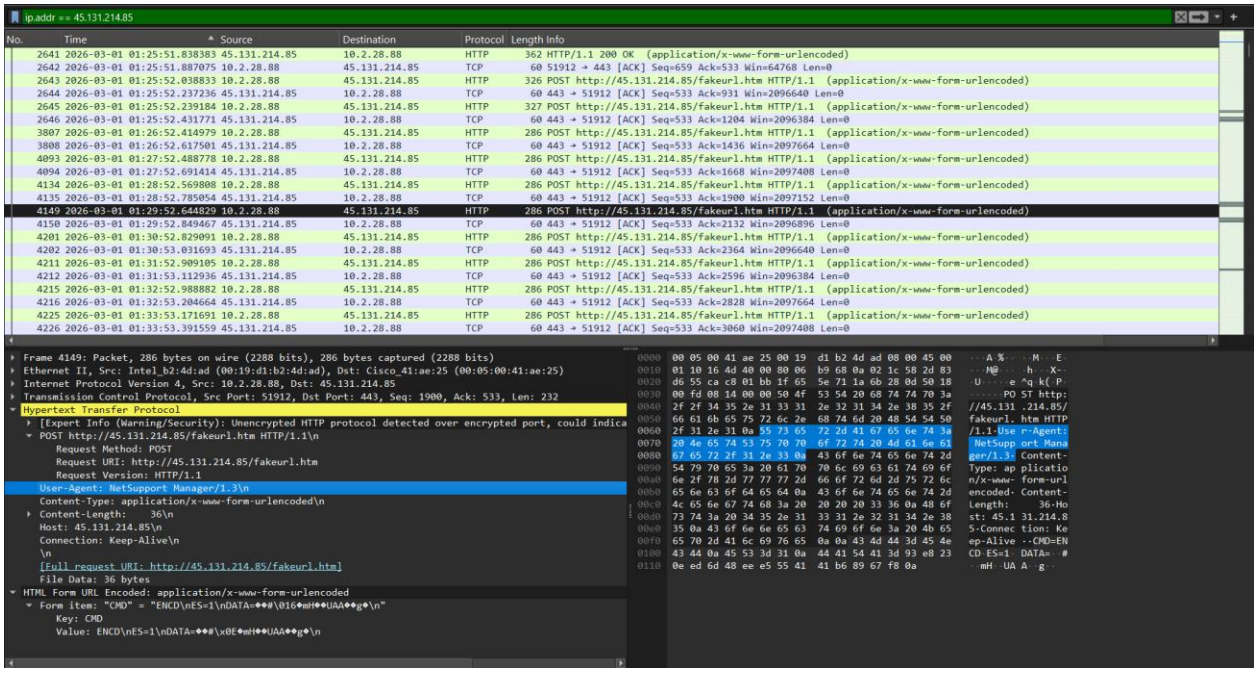


Figure: Wireshark — Detailed HTTP POST packet (ip.addr == 45.131.214.85) showing NetSupport Manager/1.3 User-Agent highlighted and ENCD encrypted payload

4. Indicators of Compromise (IOCs)

IOC Type	Value	Description
IP (External)	45.131.214[.]85	C2 server — NetSupport Manager RAT
URL	hxxp://45.131.214[.]85/fakeurl.htm	C2 beacon POST endpoint
User-Agent	NetSupport Manager/1.3	Malware HTTP User-Agent string
TCP Port	443 (unencrypted HTTP)	Port mimicry to evade firewall rules
IP (Internal)	10.2.28.88	Infected workstation — DESKTOP-TEYQ2NR
MAC Address	00:19:d1:b2:4d:ad	Infected workstation NIC
AD Account	brolf (Brad Rolf)	Logged-in user on infected host

5. Recommended Response Actions

Immediate (within 1 hour)

- Isolate DESKTOP-TEYQ2NR (10.2.28.88) — physically unplug network cable or disable switch port.
- Block 45.131.214[.]85 on the perimeter firewall and all egress points.
- Disable AD account brolf (Brad Rolf) pending investigation.
- Preserve a forensic disk image of the endpoint before any remediation.

Short-term (within 24 hours)

- Run full AV/EDR scan across all hosts in the 10.2.28.0/24 subnet.
- Review SIEM for any other internal hosts communicating with 45.131.214[.]85.
- Audit AD for recent changes to the brolf account (password resets, group membership).
- Investigate infection vector — check browser history, email, and recent downloads on the endpoint.

6. Network Environment Reference

PARAMETER	VALUE
LAN Segment	10.2.28.0/24 (10.2.28.0 – 10.2.28.255)
Domain	easyas123.tech
AD Name	EASYAS123
Domain Controller	10.2.28.2 — EASYAS123-DC
Gateway	10.2.28.1
Broadcast	10.2.28.255
pcap File	2026-02-28-traffic-analysis-exercise.pcap

Exercise Source	https://www.malware-traffic-analysis.net/